MIL-P-13981A
5 February 1971
SUPERSEDING
MIL-P-13981(CmlC)
23 February 1955

MILITARY SPECIFICATION

PHOSPHOROUS ACID, ANALYZED REAGENT

This specification is mandatory for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 This specification covers one analyzed reagent grade of phosphorous acid (H_3PO_3).

2. APPLICABLE DOCUMENTS

2.1 Government documents. The following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein.

SPECIFICATIONS

FEDERAL

PPP-B-591 - Boxes, Fiberboard, Wood-Cleated.
PPP-B-601 - Boxes, Wood, Cleated-Plywood.
PPP-B-621 - Boxes, Wood, Nailed and Lock-Corner.
PPP-B-636 - Box, Fiberboard.
PPP-C-186 - Containers, Packaging and Packing for Drugs, Chemicals, and Pharmaceuticals.
PPP-C-301 - Chemicals, Dry and Paste: Packaging and Packing of.

MILITARY

MIL-P-15011 - Pallets, Material Handling, Wood, Post Construction, 4-Way Entry.

FSC 6810

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STANDARDS

MILITARY

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129 - Marking for Shipment and Storage.
MIL-STD-147 - Palletized and Containerized Unit Loads, 40" x 48" Pallets, Skids, Runners, or Pallet-Type Base.

(Copies of specifications, standards, drawings, and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

CODE OF FEDERAL REGULATIONS

49 CFR 171-179 - Department of Transportation Rules and Regulations for the Transportation of Explosives and Other Dangerous Articles.

(The Department of Transportation regulations are a part of the Code of Federal Regulations available from the Superintendent of Documents, Government Printing Office, Washington, D. C. 20402. Orders for the above publication should cite "49 CFR 171-179.")

UNIFORM FREIGHT CLASSIFICATION RULES

(Application for copies should be addressed to the Uniform Classification Committee, 202 Union Station, Chicago, Illinois 60606.)

NATIONAL MOTOR FREIGHT CLASSIFICATION RULES

(Application for copies should be addressed to the American Trucking Association, Inc., Attention: Tariff Order Section, 1616 P Street, N. W., Washington, D. C. 20036.)

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM) STANDARDS

D1193-66 - Reagent Water.

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(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, Pennsylvania 19103.)

(Technical society and technical association specifications and standards are generally available for reference from libraries. They are also distributed among technical groups and using Federal agencies.)

3. REQUIREMENTS

3.1 Chemical characteristics. Phosphorous acid shall conform to the chemical characteristics of table I when tested as specified therein.

Table I. Chemical characteristics

Characteristic	Percent by weight		Test paragraph
	Minimum	Maximum	
Assay (as H_3PO_3)	98.0	-----	4.2.4.1
Insoluble matter	----	0.005	4.2.4.2
Chloride (Cl)	----	0.003	4.2.4.3
Phosphate (PO_4)	----	0.03	4.2.4.4
Sulfate (SO_4)	----	0.002	4.2.4.5
Heavy metals (as Pb)	----	0.002	4.2.4.6

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to specified requirements.

4.2 Quality conformance inspection.

4.2.1 Lotting. A lot shall consist of the phosphorous acid produced by one manufacturer, at one plant, from the same materials, and under essentially the same manufacturing conditions provided the operation is continuous. In the event the process is a batch operation, each batch shall constitute a lot (see 6.3).

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4.2.2 Sampling.

4.2.2.1 For examination of preparation for delivery. Sampling for examination of preparation for delivery shall be conducted in accordance with MIL-STD-105.

4.2.2.2 For test. Sampling for test shall be conducted in accordance with table II. A representative specimen of approximately 1/4 pound shall be removed from each sample unit packaging container and placed in a suitable clean, dry container labeled to identify the lot and container from which it was taken.

Table II. Sampling for test

Number of unit packaging containers in batch or lot	Number of sample unit packaging containers
2 to 25	2
26 to 150	3
151 to 1,200	5
1,201 to 7,000	8
7,001 to 20,000	10
Over 20,000	20

4.2.3 Inspection procedure.

4.2.3.1 For examination of preparation for delivery. The sample unit shall be one filled unit, intermediate, or shipping container, as applicable, ready for shipment. Sample containers and the preparation for delivery thereof shall be examined for the following defects using an AQL of 2.5 percent defective:

(a) Contents per container not as specified
(b) Container not as specified
(c) Container closure not as specified
(d) Container damaged or leaking
(e) Marking incorrect, missing, or illegible

4.2.3.2 For test. Approximately equal portions of all of the specimens taken in 4.2.2.2 shall be thoroughly mixed to form a composited specimen of no less than 1/2 pound. The composited specimen shall be tested as specified in 4.2.4. Each test shall be conducted in duplicate analysis. Failure of either analysis of any test shall be cause for rejection of the lot represented.

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4.2.4 Tests. Water in accordance with ASTM D1193 and reagent grade chemicals shall be used throughout the tests. Where applicable, blank determinations shall be run and corrections applied where significant. Tests shall be conducted as follows:

4.2.4.1 Assay (as H_3PO_3). Weigh to the nearest 0.1 milligram (mg) approximately 1 gram (g) of the specimen and dissolve in 50 milliliters (ml) of water. Titrate with approximately 1N sodium hydroxide solution using thymolphthalein as the indicator. Calculate the percent phosphorous acid (as H_3PO_3) as follows:

$$\text{Percent phosphorous acid} = \frac{4.1\ AB}{W}$$

where: A = Milliliters of sodium hydroxide solution used in the titration,
B = Normality of the sodium hydroxide solution, and
W = Weight of the specimen in grams.

4.2.4.2 Insoluble matter. Weigh to the nearest milligram approximatly 20 g of the specimen and dissolve in 20 ml of water. Filter though a tared filter crucible weighed to the nearest 0.1 mg, wash thoroughly with water, and dry the crucible and residue to constant weight at 105° ± 5° C. Cool to room temperature in a desiccator and weigh to the nearest 0.1 mg. Calculate the percent insoluble matter as follows:

$$\text{Percent insoluble matter} = \frac{100\ (A - B)}{W}$$

where: A = Weight of the crucible and residue in grams,
B = Weight of the crucible in grams, and
W = Weight of the specimen in grams.

4.2.4.3 Chloride. Weigh 1.000 g of the specimen and dissolve in 20 ml of water. Add 2 ml of concentrated nitric acid and 1 ml of 2-percent silver nitrate solution. Any turbidity produced shall be no greater than that produced when 0.03 mg of chloride ion (from sodium chloride) is treated in the same manner as the specimen.

4.2.4.4 Phosphate.

(a) Magnesia mixture. Dissolve 55 g of magnesium chloride and 70 g of ammonium chloride in 650 ml of water. Dilute to 1 liter with 10-percent ammonium hydroxide solution.

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(b) Procedure. Weigh 1.00 g of the specimen and dissolve in 20 ml of water. Add 5 ml of 10-percent ammonium hydroxide solution and 5 ml of magnesia mixture prepared as specified in (a). No turbidity shall be produced within 10 minutes.

4.2.4.5 Sulfate. Weigh 2.50 g of the specimen and dissolve in 20 ml of water. Add 2 ml of 12-percent barium chloride solution. No turbidity shall be produced within 5 minutes.

4.2.4.6 Heavy metals (as lead). Weigh 2.000 g of the specimen and dissolve in 30 ml of water. Add 10-percent ammonium hydroxide solution until the solution is faintly acid to congo red paper and then dilute to 40 ml with water. To a 10-ml aliquot, add 0.02 mg of lead ion (from lead nitrate) and dilute to 30 ml with water. This is the control solution. To both the control solution and the remaining 30 ml of specimen solution, add 2 ml of 1N acetic acid and 10 ml of freshly prepared, saturated, aqueous hydrogen sulfide solution. The color produced in the specimen solution shall be no darker than that produced in the control solution.

5. PREPARATION FOR DELIVERY

5.1 Packaging. Packaging shall be level A or C as specified (see 6.2).

5.1.1 Level A. A quantity of 1/4 pound ± 0.1 ounce of phosphorous acid shall be packaged in accordance with the level A packaging requirements of PPP-C-301 in a nominal 4-fluid ounce capacity, type I glass bottle in inclosure b as specified therein. The bottle shall also conform to group A, class 1, type optional, style 2, grade 1, with closure A and outer seal A of PPP-C-186.

5.1.1.1 Intermediate packaging. Twelve bottles of phosphorous acid, packaged as specified in 5.1.1, shall be intermediately packaged in a fiberboard box as specified in PPP-C-301 for level A intermediate packaging.

5.1.2 Level C. A quantity of 1/4 pound ± 0.1 ounce of phosphorous acid shall be packaged to afford protection against spillage, breakage, deterioration, contamination, and handling hazards from the supply source to the first receiving activity for immediate use.

5.2 Packing. Packing shall be level A, B, or C as specified (see 6.2).

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5.2.1 Level A. Four intermediate packages, prepared as specified in 5.1.1.1, shall be packed in either an overseas type, style I or J box of PPP-B-601 or a class 2, style 4-1/2 box of PPP-B-621 as specified in PPP-C-301 for level A packing.

5.2.2 Level B. Four intermediate packages, prepared as specified in 5.1.1.1, shall be packed in a box conforming to class II, style A or B of PPP-B-591; domestic type, style optional of PPP-B-601; class 1, style 4-1/2 of PPP-B-621; or style RSC, grade V3c of PPP-B-636 in accordance with the level B packing requirements of PPP-C-301 except that wood boxes, if used, need not be preserved and reinforcement strapping need not be galvanized.

5.2.3 Level C. Forty-eight bottles of phosphorous acid, packaged as specified in 5.1.2, shall be packed in containers which comply with the Code of Federal Regulations, the Department of Transportation (DOT) regulations, the Uniform Freight Classification Rules, the National Motor Freight Classification Rules, or other carrier rules as applicable to the mode of transportation. Containers shall be capable of being stacked and of supporting superimposed loads during shipment and shall assure safe delivery to destination without damage to contents. Containers shall be acceptable for shipment at the most favorable rate of the applicable regulation provided that all requirements specified herein have been met.

5.3 Marking. In addition to the special marking specified below, unit, intermediate, and shipping containers shall be marked in accordance with DOT regulations and MIL-STD-129.

5.3.1 Special marking. Each unit container shall be durably and legibly labled to show the following information:

(a) Empirical formula and formula weight
(b) Manufacturer's lot number
(c) Assay
(d) Analysis or maximum limits of impurities
(e) Precautionary marking: KEEP TIGHTLY CLOSED

5.4 Palletization. When specified in the contract or order (see 6.2), shipping containers shall be palletized in accordance with the applicable requirements of MIL-STD-147 using a pallet conforming to style 1, type I, class 1 of MIL-P-15011. The pallet and other wood storage aids, if used, shall be treated with wood preservative as specified for the pallet in MIL-P-15011.

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6. NOTES

6.1 Intended use. Phosphorous acid is intended for use as a laboratory reagent.

6.2 Ordering data. Procurement documents should specify the following:

(a) Title, number, and date of this specification.
(b) Level of packaging and packing required (see 5.1 and 5.2).
(c) Palletization, if required (see 5.4).

6.3 Batch. A batch is defined as that quantity of material which has been manufactured by some unit chemical process or subjected to some physical mixing operation intended to make the final product substantially uniform.

6.4 Significant places. For the purpose of determining conformance with this specification, an observed or calculated value should be rounded off "to the nearest unit" in the last right-hand place of figures used in expressing the limiting value, in accordance with the rounding-off method of the Recommended Practices for Designating Significant Places in Specified Limiting Values (ASTM E29).

Custodians:

Army - MU
Air Force - 68

Review activities:

Army - MD, MU(FA)
DSA - GS

User activities:

Army - MI, SM

Preparing activity:

Army - MU(EA)

Project No. 6810-0611

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SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 22-R255

INSTRUCTIONS: This sheet is to be filled out by personnel, either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity. Comments and suggestions submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or serve to amend contractual requirements.

SPECIFICATION

ORGANIZATION

CITY AND STATE

CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO *(If "yes", in what way?)*

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity - Optional)*

DATE

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OFFICIAL BUSINESS

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